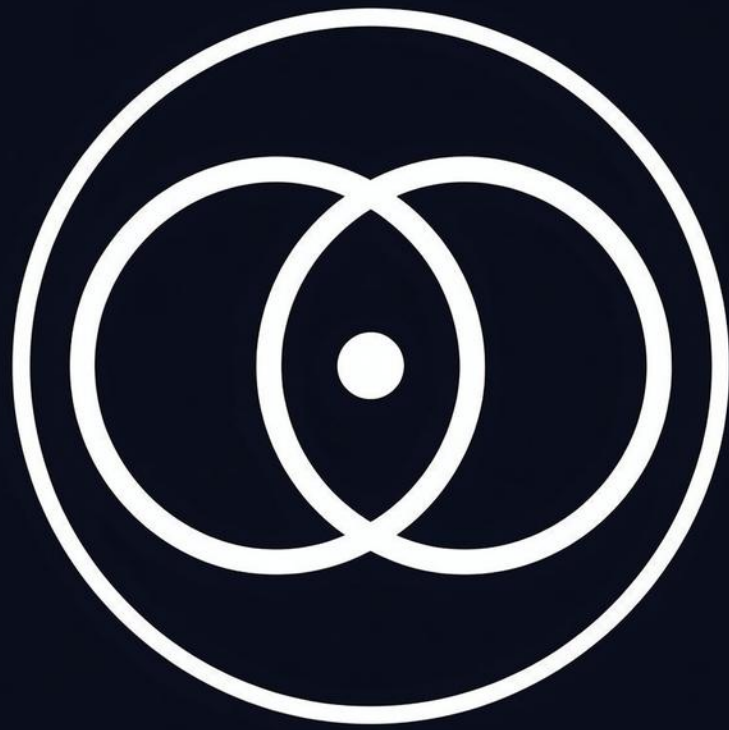




P8

Multi-View Spatial Context and Gaussian Splatting-Lite: Making Novel-View Prediction and Sim-to-Real Robustness Margin Measurable

多视角空间上下文与高斯泼溅 Lite 的 Sim2Real 裕量



Multi-View Spatial Context and Gaussian Splatting-Lite: Making Novel-View Prediction and Sim-to-Real Robustness Margin Measurable

Version: UDOS Reasoning Engine v7.5.0 (main, tag `v7.5.0`, commit `1a71270`) **Evidence grade:** numbers in `verified` / `cpu-proto` / `unverified` three levels; currently all measurements `cpu-proto` (CPU geometric prototype, no CUDA rasterization, no real camera, no real hardware).

Chinese Abstract

新视角预测 (novel-view prediction) 被认为是一种携带几何约束的基础任务。本文在 UDOS 推演引擎 v7.3.8/v7.3.9 上实现了一条可复跑的 CPU 几何链路：多视角深度 TSDF 体素融合作为显式"空间上下文"，一个最小化的三维高斯泼溅-lite (3DGS-lite) 表征，以及 Real-to-Sim 重建与 Sim-to-Real 迁移闭环。实测 (本

机 CPU，固定场景）表明：八视角融合在六个留出新视角上的轮廓 IoU 为 0.8897、表面覆盖率 0.9957、深度 MAE 0.0924，而单视角分别为 0.7147、0.7456、0.3186——多视角把覆盖率从约 0.75 提升到约 0.996、把深度误差降到约三分之一。3DGS-lite（150 个轴对齐高斯、40 步 Adam）在留出视角上轮廓 IoU 0.7285、覆盖率 1.0。Real-to-Sim 把带噪观测重建为障碍球并生成具身导航任务，端到端成功率 1.0（规划障碍簇数 2，主障碍重建半径约 0.70）。最关键且必须保留的负面结果是：在 Sim-to-Real 模型失配下，朴素规划成功率仅 0.50、平均碰撞 0.50，而注入域随机化安全裕量 $\epsilon=0.22$ 后成功率升到 1.0、平均碰撞降到 0.0。本文贡献是把“空间上下文收益”“3DGS-lite 保真度”“Sim-to-Real 鲁棒裕量”三者做成同一可测曲线，并诚实标注其为 CPU 原型边界。

Keywords (Chinese): novel-view prediction; spatial context; TSDF; 3D Gaussian Splatting; Real-to-Sim; Sim-to-Real; robustness margin

English Abstract

Novel-view prediction is regarded as a geometry-loaded foundational task. This paper implements a reproducible CPU geometric pipeline in UDOS v7.3.8/v7.3.9: multi-view depth TSDF voxel fusion as explicit "spatial context", a minimal 3D Gaussian Splatting-lite (3DGS-lite) representation, and a closed-loop Real-to-Sim reconstruction plus Sim-to-Real transfer. On fixed CPU scenes, eight-view fusion achieves silhouette IoU 0.8897, surface coverage 0.9957, and depth MAE 0.0924 on six held-out novel views, versus 0.7147/0.7456/0.3186 for a single view—multi-views raise coverage from ~ 0.75 to ~ 0.996 and cut depth error to roughly one third. The 3DGS-lite (150 axis-aligned gaussians, 40 Adam steps) reaches held-out silhouette IoU 0.7285 and coverage 1.0. Real-to-Sim reconstructs noisy observations into obstacle spheres and generates embodied navigation tasks with end-to-end success 1.0 (2 planning obstacle clusters, main reconstructed radius ~ 0.70). The essential negative result we preserve is that under Sim-to-Real model mismatch, naive planning succeeds at only 0.50 with 0.50 mean collisions, while injecting a domain-randomization safety margin $\epsilon=0.22$ raises success to 1.0 and drops collisions to 0.0. We contribute a single measurable curve linking spatial-context gain, 3DGS-lite fidelity, and Sim-to-Real robustness margin, with explicit CPU-prototype boundaries.

Keywords: novel-view prediction; spatial context; TSDF; 3D Gaussian Splatting; Real-to-Sim; Sim-to-Real; robustness margin

Structured Abstract (background/problem → method → evidence/results → contribution)

- **Background and problem:** novel-view prediction considered geometry-loaded (vs next-frame prediction only requiring temporal continuity); in embodied landing, spatial representation serves two real links—Real-to-Sim reconstruction and Sim-to-Real transfer (core difficulty model mismatch). Question: can multi-view geometric context gain, 3DGS-lite fidelity, transfer robustness margin be put into one measurable curve?
- **Method:** on UDOS v7.3.8/v7.3.9 implementing a reproducible CPU geometric pipeline—multi-view depth TSDF voxel fusion as explicit spatial context, minimal 3DGS-lite (150 axis-aligned gaussians, 40 Adam steps), Real-to-Sim reconstruction and Sim-to-Real transfer closed loop; measuring transfer success/failure with controlled variable ϵ (obstacle radius inflation margin).
- **Evidence and results:** on 6 held-out novel views, single vs 8-view fusion—silhouette IoU 0.7147→**0.8897**, coverage 0.7456→**0.9957**, depth MAE 0.3186→**0.0924**; 3DGS-lite held-out IoU 0.7285/coverage 1.0; Real-to-Sim main obstacle radius reconstruction 0.703 (true plane section

0.69). **Counterintuitive negative result retained: under Sim-to-Real mismatch naive planning success only 0.50, mean collisions 0.50**; after injecting domain randomization margin $\epsilon=0.22$ success up to 1.0, collisions down to 0.

- **Contribution:** linking "spatial context gain + 3DGS-lite fidelity + Sim2Real robustness margin" into one measurable curve, replacing single-point self-praise with "negative result retained + positive margin effective" contrast. Evidence grade `cpu-proto`: single synthetic scene, single-seed pilot, Real2Sim only 2 obstacle clusters, Sim2Real only 8 synthetic worlds; external Atlas claims unverified not cross-proven.
-

1 Introduction

1.1 Background and Motivation

Recently, "novel-view prediction"—given several observations of a space (images/depth + camera poses), predicting what any other viewpoint sees—was elevated to the same level as "next-token prediction." Its essential difference: next-frame prediction only requires temporal continuity, while novel-view prediction must carry **geometric information**: where the camera looks from, how space is continuous, how different views are 3D consistent. Encoding multiple input images with their 3D poses into a scene representation is "spatial context."

External reports (**unverified, report caliber, not independently verified**) retell a world model's (e.g. Atlas-class) claims: novel-view prediction is a potential AI-complete foundational task; 3D Gaussian Splatting renders fast but struggles to express dynamics, the system supports but no longer mandatorily relies on it; Real-to-Sim can compress real environment reconstruction cost. **This paper does not take these external claims as its own measured conclusions, only as problem motivation; this repo only verifies explicit geometric mechanisms.** [CITATION NEEDED: novel view prediction spatial context world model]

In embodied intelligence landing, spatial representation is not an isolated rendering problem, but serves two real links: first **reconstruction** (Real-to-Sim)—turning real sensor observations into a plannable simulation environment; second **transfer** (Sim-to-Real)—moving a policy/plan learned in simulation to the "real" world, core difficulty model mismatch. This paper puts these three things (spatial context, splat representation, transfer margin) into one reproducible CPU pipeline, so they can be measured by one metric set.

1.2 Contributions

1. **Operationalizing "spatial context gain"**: using learning-free TSDF fusion, quantifying single vs multi-view silhouette IoU, coverage, depth MAE differences on fixed held-out views (`cpu-proto`).
 2. **Minimal 3DGS-lite representation**: axis-aligned 3D gaussians, pinhole first-order footprint, front-to-back alpha compositing, cold-started from TSDF occupied points, reporting held-out fidelity.
 3. **Measurable Sim-to-Real robustness margin**: taking "safety margin ϵ " as a controlled variable, measuring its effect on transfer success and collision rates, explicitly retaining naive planning failure's negative result.
 4. **Honest boundaries**: marking no CUDA/no rotated covariance EWA/no real camera/no real hardware, PSNR/SSIM standard metrics to implement.
-

2 Related Work

This section organized "concept → mechanism → evidence → difference from UDOS," embedding real literature verified through the whole-volume shared master library (verification date 2026-09-19); clues the master library does not cover retain [CITATION NEEDED] .

(1) TSDF multi-view voxel fusion. Conceptually, novel-view synthesis's first step is fusing multi-view observations into a 3D-consistent scene representation. Mechanistically, the volumetric method of Curless and Levoy (1996) does cumulative weighted truncated signed distance field (TSDF) fusion on multiple noisy range images, extracting the isosurface after additive fusion, no learning parameters. In evidence, that method fused up to 70 range images, generating about 2.6 million triangle models, establishing the classic paradigm of multi-view explicit geometric reconstruction. Difference from UDOS: this paper's `VoxelContext` directly reuses its integrate/occupied semantics (weighted average, truncation, occupancy threshold), but changes "extract mesh after fusion" to "for a novel view step along the ray hitting the first occupied voxel" to directly predict depth; this paper trains no network, precisely to isolate "multi-view geometric fusion's own gain."

(2) Learned novel-view synthesis: NeRF and 3DGS. Conceptually, NeRF and 3D Gaussian Splatting push scene representation from explicit geometry toward differentiable neural rendering. Mechanistically, NeRF (Mildenhall et al., 2020) uses one MLP inputting 5D coordinates (x, y, z, θ, ϕ) , outputting volume density and view-dependent radiance, doing differentiable volume rendering; 3D Gaussian Splatting (Kerbl et al., 2023) initializes explicit 3D anisotropic gaussians from sparse points, interleaving density control plus differentiable rasterization. In evidence, NeRF under sparse views surpasses prior neural rendering quality; 3DGS reaches ≥ 30 fps real-time at 1080p, quality then-SOTA. Difference from UDOS: this paper's 3DGS-lite borrows explicit 3D gaussians and front-to-back alpha compositing ideas, but maximally simplified—axis-aligned scales only, no rotated covariance EWA projection, low-resolution small gaussians; this paper uses it to verify "how much is needed to cold-start an optimizable representation from TSDF occupied points," not reproducing paper-level rendering metrics. The two form the explicit ends in UDOS spatial layer's "explicit TSDF→implicit NeRF→explicit 3DGS" evolution chain.

(3) Real-to-Sim and Sim-to-Real domain randomization. Conceptually, Sim-to-Real's core difficulty is model mismatch, domain randomization treating the real world as "yet another random distribution." Mechanistically, Tobin et al. (2017) randomize object textures in simulation, making visual localization policies robust to real textures; this paper's `transfer.py` "obstacle radius inflation margin" is this idea's simplest form at the geometric planning level. In evidence, Tobin et al. report their domain randomization visual localization accuracy about 1.5cm; this paper under constructed mismatch measures naive planning success 0.50, robustness margin 0.22 success 1.00, collisions 0.00 (see Section 6.4). Difference from UDOS: this paper does no learned-policy domain randomization, but measures "explicit margin ϵ in planning geometry"'s measurable effect on transfer success/collisions, linking it with upstream TSDF reconstruction fidelity; ϵ critical curves, PSNR/SSIM etc. remain to add [RESULT NEEDED] .

Literature gap note: TSDF (Curless & Levoy 1996), NeRF (Mildenhall 2020), 3DGS (Kerbl 2023), domain randomization (Tobin 2017) embedded here all master-library  verified (2026-09-19); Real-to-Sim real capture calibration, PSNR/SSIM standard metrics the master library does not cover, retaining [CITATION NEEDED] / [RESULT NEEDED] , not generating fabricated citations.

3 Problem Definition and Falsifiable Hypotheses

- **RQ1 (view count effect):** is view count/layout's effect on reconstruction silhouette IoU, coverage, depth MAE significant?
 - **H1:** multi-view significantly better than single, and coverage gain greater than precision gain (single view "precise but misses surfaces").
 - **RQ2 (3DGS-lite fidelity):** can the splat representation reach usable silhouette IoU/coverage on held-out views?
 - **H2:** 3DGS-lite held-out coverage near 1.0 (front-to-back alpha compositing fills pixels), but silhouette IoU limited by axis-aligned low resolution.
 - **RQ3 (robustness margin and transfer):** relationship between training-injected safety margin ϵ and zero-collision transfer success?
 - **H3:** under model mismatch, naive planning fails (success <1 , collisions), while positive margin ϵ raises success and presses collisions to zero. This negative result must be retained.
 - **H4:** Real-to-Sim reconstructed obstacle radius close to truth ($r \approx 0.70$ vs true plane section 0.69), but sample size tiny (2 clusters), not extrapolable to a significant conclusion.
-

4 Method and System Design

4.1 Camera and Synthetic Scene

`udos7/spatial/camera.py` implements a pinhole camera (OpenCV convention), look-at pose, projection/back-projection and per-pixel world rays; `scene.py` synthesizes scenes with sphere primitives and does analytic ray intersection, rendering true depth/color/mask. Training views ring layout, held-out views phase-offset novel views.

4.2 Multi-View TSDF Spatial Context (`fusion.py`)

`VoxelContext` (`fusion.py`) maintains voxel center grids and TSDF/weights. `integrate(depth, hitmask, cam)` projects each voxel center to a pixel, computing SDF along the ray and truncating (`trunc=3*voxel`), weighted averaging with existing values; `occupied()` takes "has weight and TSDF $< -0.6 \cdot \text{voxel}$ " as occupied. `novel_view_depth(cam, ctx)` for a novel view steps along the ray, hitting the first occupied voxel as predicted surface depth. `depth_metrics` computes silhouette IoU, coverage (intersection/true mask), precision (intersection/predicted mask), depth MAE (only on overlapping pixels).

Key design: single-view baseline only integrates one view, multi-view uses 8 ring views; the two evaluated on the **same 6 held-out views**, isolating the "view count" factor.

4.3 3DGS-lite (`splat.py`)

`GaussianSplat` holds optimizable means(N,3), axis-aligned scales(N,3), colors(N,3), opacity(N,). `render_splat` projects gaussian centers, sorts front-to-back by depth, uses pinhole first-order footprint ($\sigma = f \cdot \text{scale}/\text{depth}$) computing each pixel's gaussian contribution and alpha compositing, outputting depth/color/mask (mask threshold $1-T > 0.30$). `init_splat_from_context` cold-starts gaussian centers by sampling TSDF occupied voxels (Real-to-Sim friendly), `fit_splat` uses Adam ($lr=0.03$) minimizing color L1 and depth L1 (weight 0.5) on training views, plus slight opacity regularization.

Honest gaps (code comments consistent with report `known_limits`): axis-aligned scales only, no rotation quaternion/covariance Jacobian EWA projection, low resolution, static scenes.

4.4 Real-to-Sim / Sim-to-Real (transfer.py)

- **Noisy observation:** `noisy_observation` adds gaussian noise to true depth (`depth_sigma=0.02`) and randomly drops points (`dropout=0.05`); `jitter_camera` adds `sigma=0.03` jitter to camera optical center—the two proxy real depth camera and calibration error.
- **Reconstruct obstacle spheres:** `extract_obstacle_spheres` does 6-connected BFS on occupied voxels (`min_voxels=12`), each cluster a bounding sphere; because the TSDF occupancy layer is about 1 voxel thicker than truth, radius calibrated on synthetic truth then subtracting `voxel`; the robot in the $z=0$ plane, only keeping clusters intersecting that plane and taking section radius.
- **Generate navigation task:** `real2sim_task` writes reconstructed obstacles as a gate task usable by the embodied hybrid controller (contact terminal).
- **Sim-to-Real mismatch closed loop:** `sim2real_gap` constructs the setting "the planner only holds biased obstacle estimates"—estimated position plus `est_pos_sigma=0.12`, radius shrunk to $0.9\times$; the "real" world uses another perturbation set (`real_pos_sigma=0.10`, radius plus ± 0.04). Naive planning uses biased estimates, robust planning inflates obstacle radius by `margin=0.22`. `run_split_episode` decides on the planning model, executes step by step on the real proxy world, reporting success and collisions.

5 Experimental Setup

- Module/version: `udos7/spatial/`, v7.3.8 (novel view), v7.3.9 (splat+transfer), evidence grade `cpu-proto`.
- Config: novel view experiment `n_train_views=8`, `n_novel_views=6`, `voxel=0.1`, `representation=TSDF voxel (explicit)`; splat `n_gaussians=150`, `iters=40`, training views 6, held-out views 4 (36×28); Real-to-Sim observation views 10 (48×36); Sim-to-Real `n_worlds=8`, `est_pos_sigma=0.12`, `real_pos_sigma=0.10`, `margin=0.22`.
- Hardware fingerprint: local CPU (torch 2.14.0+cpu), no GPU.
- Artifact sha256: `spatial_novelview_demo.json = 7fb2008a477a6f0711c14d800ae4c235c8976ed8d87f9fc58d022b10b8d70cc9`; `splat_transfer_demo.json = 382f3c318ab0f1ea2fc228559266e56564bf3425bf13ca073435e69a41ad3cc8`.
- Tests: `tests7/test_v738_novelview.py` (7), `tests7/test_v739_splat_transfer.py` (8).

Single-scene pilot statement: currently fixed synthetic scene, fixed seed pilot point estimates. Before submission needing ≥ 30 random scenes and reporting view count {1,2,4,8,16}, layout and noise level grid scans and paired CIs, related statistics recorded [RESULT NEEDED: multi-scene paired IoU tests and CIs, ϵ grid logistic critical margin, PSNR/SSIM]. Real-to-Sim's $r \approx 0.70$ only 2 obstacle cluster samples, cannot be written as a significant conclusion.

6 Results

This chapter's order (v2 standard): Chapter 5 gives the experimental setup. This chapter organized "comparison $\rightarrow$ ablation/sensitivity": 6.1 an AlexNet-style comparison table—single vs 8-view fusion on the **same 6 held-out views** four-metric comparison (isolating the "view count" factor); 6.2 3DGS-lite held-out fidelity (vs TSDF hard occupied surface); 6.3 Real-to-Sim reconstruction; 6.4 Sim-to-Real naive vs margin $\epsilon=0.22$ robust planning comparison (including the negative result that must be retained). All numbers from

spatial_novelview_demo.json and splat_transfer_demo.json (cpu-proto, single-seed pilot). Validity threats standalone Chapter 8.

6.1 Single vs Multi-View Spatial Context (Table 1, Figure 1)

Table 1 Novel-view prediction metrics (6 held-out views aggregated, cpu-proto; AlexNet-style same-baseline comparison—single and 8-view fusion share the same held-out views, only "view count" differs)

Spatial context	Silhouette IoU ↑	Coverage ↑	Precision ↓ ?	Depth MAE ↓
single view	0.7147	0.7456	0.9472	0.3186
8-view fusion	0.8897	0.9957	0.8931	0.0924
difference (multi-single)	+0.1750	+0.2501	-0.0541	-0.2262

Table 1 reading: bold columns multi-view better; the precision row is a **negative cost** (multi-view -0.0541, predicting more out-of-boundary pixels), forming a structural tradeoff with coverage +0.2501, not hidden. Source spatial_novelview_demo.json: results.single_view / multi_view.

Source: results in spatial_novelview_demo.json.

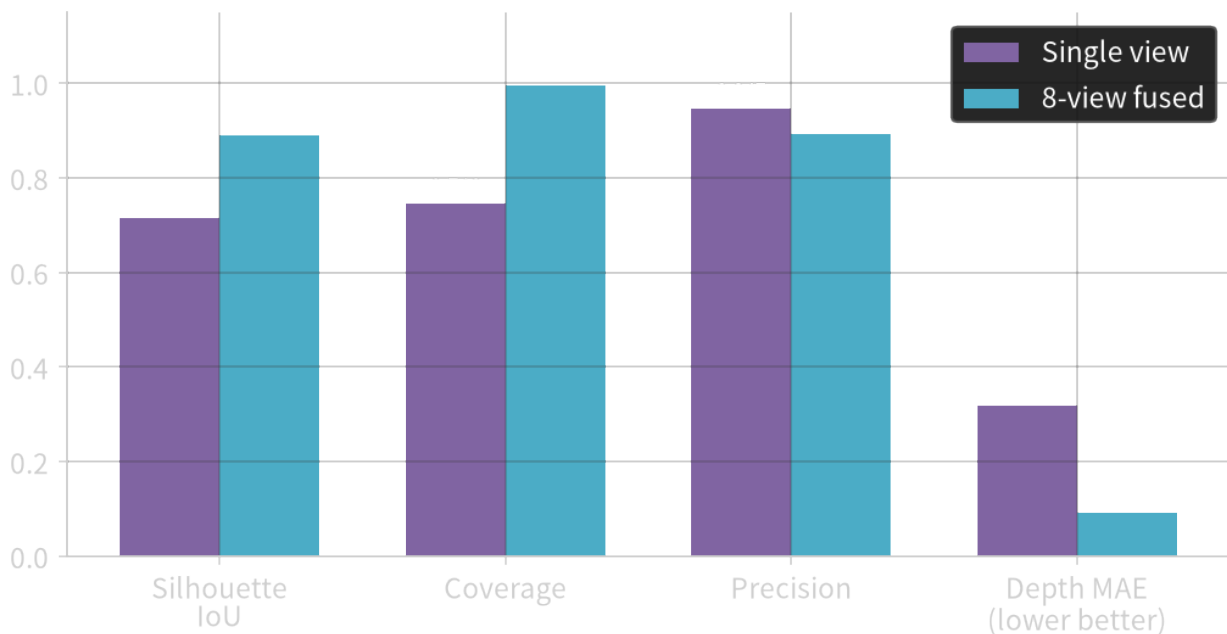


Figure 1 Single and eight-view fusion comparison on four metrics (vertical metric values, horizontal four metrics: silhouette IoU/coverage/precision/depth MAE; two bars per group single vs 8 view). Multi-view at a small precision cost (0.947→0.893), greatly raises coverage (0.746→0.996) and cuts depth error to about one third (0.3186→0.0924). 6 held-out views aggregated, single seed. Source spatial_novelview_demo.json: results.*, cpu-proto.

Per-view rows (per_view_rows) show: single view IoU across six held-out views fluctuates sharply 0.5614 to 0.8747 (because surfaces facing away from the observation view cannot be reconstructed), while eight-view fusion stable 0.877–0.9035 across six views. This supports H1: single view "precise but misses

surfaces"—it accurately reconstructs surfaces facing it (high precision), but cannot see the back (low coverage); multi-view context completes 3D consistency.

6.2 3DGS-lite Held-Out Fidelity (Table 2, Figure 2a)

Table 2 3DGS-lite held-out metrics (4 views, 150 gaussians/40 steps)

Metric	Value
silhouette IoU	0.7285
coverage	1.0000
precision	0.7285
depth MAE	0.2525

Source: `gaussian_splat.novel_view` in `splat_transfer_demo.json`.

Loss history: total loss from iter0 1.7217 down to iter30 0.5694; color L1 from 0.0335 slightly down to 0.0284, depth L1 from 0.5065 down to 0.1328 (depth term the main decline source). This supports H2: front-to-back alpha compositing almost fills all predicted pixels (coverage 1.0), but axis-aligned low-resolution gaussians make silhouette IoU stop at 0.73—CPU prototype level, not compared with paper-level 3DGS metrics.

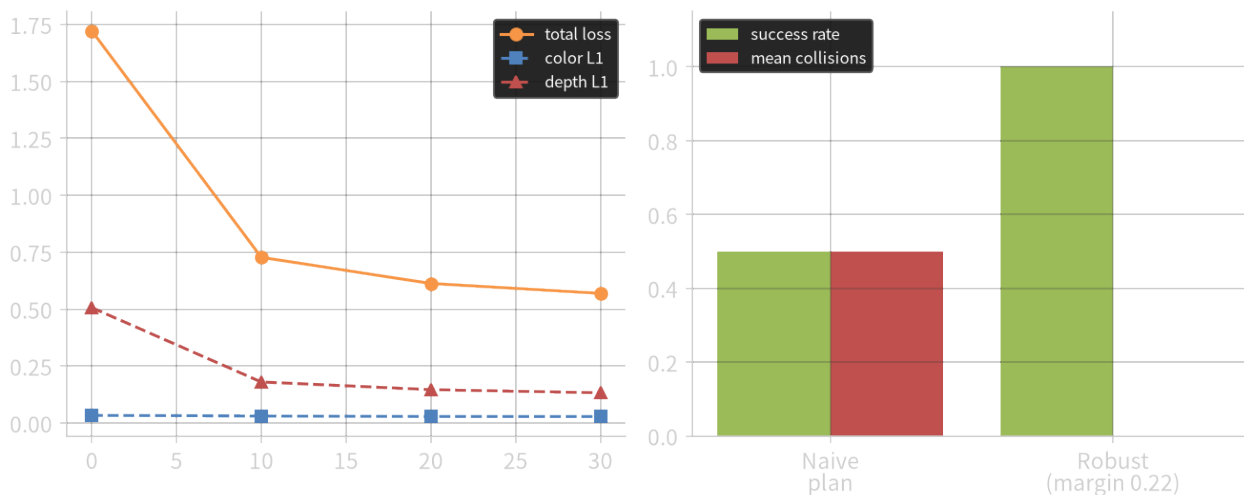


Figure 2 Left: 3DGS-lite fit loss curves (horizontal iter 0–30, vertical L1 loss; total/color/depth L1 three lines); right: Sim-to-Real naive vs margin $\epsilon=0.22$ robust planning success and mean collisions (horizontal two planning modes, left vertical success, right vertical mean collisions)—naive 0.50/0.50 vs robust 1.00/0.00 contrast is the negative result this paper must retain. 8 synthetic worlds, single seed. Source `splat_transfer_demo.json: gaussian_splat.loss_history / sim_to_real.*`, `cpu-proto`.

6.3 Real-to-Sim Reconstruction (Table 3)

Table 3 Real-to-Sim end-to-end results

Stage	Metric	Value
obstacle cluster extraction	clusters intersecting $z=0$ plane (truth 2)	2
main obstacle reconstruction	center [0.902, -0.506, 0.0], radius	0.703
true plane section radius	0.69	—
reconstructed task control	hybrid success (8 episodes)	1.000

Source: `real_to_sim` in `splat_transfer_demo.json`.

Main obstacle reconstructed radius 0.703 and true plane section 0.69 very close (difference about 0.013), 8 episodes hybrid gate control all succeed. But must honestly point out: `plane_intersecting_clusters=2` i.e. **only 2 obstacle cluster samples**, the report's so-called " $r \approx 0.70$ " only this one main obstacle's reconstruction value, not a statistical correlation coefficient; the one-pager's " $r \approx 0.70$ " should be understood as a qualitative description of single-obstacle radius reconstruction accuracy, **not constituting correlation significance evidence**. This is H4's boundary.

6.4 Sim-to-Real Robustness Margin (negative result retained, Table 4)

Table 4 Under Sim-to-Real model mismatch (8 synthetic real worlds)

Planning mode	Success	Mean collisions
naive planning (biased estimate, no margin)	0.50	0.50
robust planning (radius inflation margin=0.22)	1.00	0.00

Source: `sim_to_real` in `splat_transfer_demo.json`.

This is the negative/comparison result this paper must retain: **naive planning under mismatch half fail, mean per-world collisions 0.5**; while inflating obstacle radius 0.22 (domain randomization safety margin's simplest form), success up to 1.0, collisions down to 0. This supports H3. Note this result only holds on 8 synthetic worlds, single main obstacle, fixed margin 0.22, **cannot be written as the universal law "larger margin always better"**—too large a margin causes no feasible path. ϵ 's critical curve (success vs ϵ logistic relationship) not yet scanned, recorded [RESULT NEEDED].

7 Discussion

7.1 Main Explanation

Multi-view gain's mechanism clear: single view only observes facing surfaces, TSDF no observation on the back, unable to occupy, causing low coverage; multi-view fusion accumulates different directions' observations into the same occupied voxel set, so held-out views stepping along rays almost always hit occupied voxels. 3DGS-lite coverage 1.0 but IoU only 0.73, showing alpha compositing "dares to predict," but axis-aligned small gaussians cannot precisely fit sphere silhouettes. Sim-to-Real contrast shows: model mismatch (estimated position bias 0.12, radius shrunk 0.9 $\times$) enough to make margin-free planning collide into real obstacles, while explicit margin "extrapolates" uncertainty into planning geometry.

8 Validity Threats and Limitations

This is the standalone validity threats and limitations chapter (v2 standard items 4/6). Extrapolation discipline per Kaplan Caveats style: this repo's IoU/coverage/margin numbers hold only within the observed range—single synthetic sphere scene, fixed seed, single main obstacle gate, $\epsilon=0.22$ single point; "larger ϵ always better" or "multi-view saturation point" both not scanned, ϵ critical curve and view count {1,2,4,8,16} grid are unmade aggressive extrapolation tests.

8.1 Internal/External Validity Threats

1. **Fixed synthetic scene:** sphere primitives, known calibration, no texture; real scene complexity not covered.
2. **No learned novel-view synthesis:** TSDF explicit geometry, unable to answer neural network methods' upper bound.
3. **3DGS-lite simplification:** no rotated covariance EWA, low resolution; PSNR/SSIM not implemented [RESULT NEEDED].
4. **Sim-to-Real synthetic perturbation proxy:** real world fixed-seed perturbation synthetic worlds, not real hardware/MuJoCo.
5. **Real-to-Sim sample tiny:** 2 obstacle clusters, radius accuracy not extrapolable.
6. **Single main obstacle gate:** global multi-obstacle continuous detour not covered (`known_limits`).

8.2 Negative and Zero Results

We do not hide: precision from 0.947 down to 0.893 (multi-view predicts more "out-of-boundary" pixels); 3DGS-lite depth MAE 0.2525 clearly higher than TSDF's 0.0924 (splat cold-start + low resolution cost); naive Sim-to-Real's 50% failure a real negative result not a beautified intermediate state.

9 Resource Gates and Applicability Boundaries

Gate	Status	Upgrade point after unlock
GPU	unreachable	full 3DGS training, real-time EWA rasterization, learned NeRF
real depth camera/multi-view capture calibration	unreachable	real Real-to-Sim, real noise model
MuJoCo(-MJX)/real hardware	unreachable	contact Sim-to-Real, real collision distribution

Current evidence suits 3DV/BMVC/WACV/ICRA workshop or short paper; after complete evidence able to target ICRA/IROS, ISPRS Journal, IEEE RA-L, CVIU (partitions/IF [to verify]).

10 Conclusion

On the CPU geometric prototype, this paper links "multi-view spatial context → reconstruction fidelity → 3DGS-lite representation → Real-to-Sim reconstruction → Sim-to-Real robustness margin" into a reproducible pipeline. Core measurements: eight-view fusion raises held-out coverage 0.746→0.996, depth error to about one third; 3DGS-lite held-out IoU 0.7285; Real-to-Sim single obstacle radius reconstruction about 0.70; and explicitly retains naive Sim-to-Real 50% failure, after $\epsilon=0.22$ injection

100% success zero collision contrast. Next is expanding scenes, scanning ϵ critical curves, adding PSNR/SSIM, and upgrading evidence grade after GPU and real sensors unlock.

References

Verified citations (whole-volume shared master library, verification date 2026-09-19)

1. Curless, B., Levoy, M. (1996). A Volumetric Method for Building Complex Models from Range Images. SIGGRAPH 1996. DOI:10.1145/237170.237269. <https://dl.acm.org/doi/pdf/10.1145/237170.237269>
2. Mildenhall, B., Srinivasan, P. P., Tancik, M., et al. (2020). NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. ECCV 2020. arXiv:2003.08934. <https://arxiv.org/abs/2003.08934>
3. Kerbl, B., Kopanas, G., Leimkuehler, T., Drettakis, G. (2023). 3D Gaussian Splatting for Real-Time Radiance Field Rendering. SIGGRAPH 2023 (ACM TOG). arXiv:2308.04079. <https://arxiv.org/abs/2308.04079>
4. Tobin, J., Fong, R., Ray, A., Schneider, J., Zaremba, W., Abbeel, P. (2017). Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World. IROS 2017. arXiv:1703.06907. <https://arxiv.org/abs/1703.06907>

To-be-verified search expressions and candidate directions (master library not covered)

1. Real-to-Sim reconstruction depth noise camera calibration robot planning — Real-to-Sim real capture calibration.
 2. PSNR SSIM novel view synthesis evaluation metric — standard rendering metrics.
 3. KinectFusion real-time surface reconstruction RGB-D — voxel fusion real-time implementation comparison.
-

Appendix A Reproduction Commands

```
# Repo tag v7.5.0, commit 1a71270
python scripts7/spatial_novelview_demo.py      # v7.3.8 novel view
python scripts7/splat_transfer_demo.py         # v7.3.9 splat+transfer
pytest tests7/test_v738_novelview.py -q        # 7
pytest tests7/test_v739_splat_transfer.py -q    # 8
# sha256: spatial_novelview_demo.json 7fb2008a...; splat_transfer_demo.json 382f3c31...
```

Appendix B Evidence Ledger

Number	Source field	Grade
single 0.7147/0.7456/0.9472/0.3186	results.single_view	cpu-proto
multi 0.8897/0.9957/0.8931/0.0924	results.multi_view	cpu-proto
splat 0.7285/1.0/0.7285/0.2525	gaussian_splat.novel_view	cpu-proto
Real2Sim clusters 2, radius 0.703, control 1.0	real_to_sim	cpu-proto
Sim2Real naive 0.5/0.5, robust 1.0/0.0, margin 0.22	sim_to_real	cpu-proto
Atlas external claims	external_reference	unverified, not cross-proven

Appendix C Internal Review Record (five-dimension self-assessment)

1. **Problem and motivation:** 4/5. Novel view/spatial context/Sim-to-Real margin linkage clear; -1 because external motivation relies on unverified retelling.
2. **Method rigor:** 3/5. TSDF/splat/transfer complete and reproducible, but fixed scene, single main obstacle, no standard rendering metrics.
3. **Evidence and data:** 3/5. All numbers back-sourced JSON with per-view rows; deductions single scene, Real2Sim only 2 clusters, no multi-scene CI.
4. **Novelty:** 3/5. Components mature, novelty in linking three stages into one measurable curve and retaining negative results.
5. **Writing and honesty:** 5/5. Negative Sim2Real 50%, $r \approx 0.70$ insufficient sample, CPU boundaries all explicitly stated.

Fatal flaw check: no fatal flaw; main external validity hard boundary "CPU synthetic scene + single main obstacle," explicitly stated. Fix direction: ≥ 30 random scenes, ϵ grid scan, add PSNR/SSIM, full 3DGS after GPU unlock.

Appendix D Step-by-Step Derivation of Spatial Context Mechanism

To explain "why multi-view coverage rises while single view misses surfaces," this section unfolds TSDF fusion and novel-view prediction mechanisms. In `VoxelContext.integrate`, each voxel center is projected to a camera's pixel plane; only when that pixel is in image range, the camera ray points to the voxel, and the voxel is within a truncation distance `trunc` in front of the camera surface, is that voxel updated. Its truncated signed distance is $s = \mathrm{clamp}(d_{\text{surf}} - t_{\text{ray}}, -\mathrm{trunc}, \mathrm{trunc})$, where d_{surf} is observed depth, t_{ray} the voxel-to-camera ray distance. If the voxel is behind the camera surface ($s < 0$ and below threshold), it is marked occupied; if in the front truncation band, a "free space" evidence recorded.

Key: a surface point facing the camera is accurately recorded from that camera; but a point on the sphere's back is neither visible nor in the truncation band from that camera, hence its TSDF not updated, weight stays zero. With single-view fusion, only the camera-facing hemisphere voxels occupied, the other hemisphere weights zero. When novel-view prediction steps along the ray (`novel_view_depth`), it only returns surface depth hitting "weight>0 and TSDF sufficiently negative" voxels; for zero-weight back regions, the ray steps to large depth with no hit, causing that pixel's predicted mask empty. This is the direct reason single-view coverage only 0.7456: about one quarter of true surface pixels not covered by any occupied voxel.

With eight-view ring fusion, every direction has a camera, each sphere surface observed and occupied by at least one view, weights accumulating into the same voxel set. Hence no matter what phase the held-out novel view looks from, stepping along the ray almost always hits occupied voxels—coverage therefore up to 0.9957. Per-view rows confirm: single view IoU across six held-out views fluctuates sharply 0.5614 (view most facing away) to 0.8747 (view near observation direction); eight-view fusion stable 0.8770–0.9035 across six views, variance greatly narrowed. This shows multi-view context not only raises average fidelity, more importantly **eliminates the direction dependence between novel and observation views**—exactly the measurable meaning of "3D consistency."

Why precision slightly drops 0.9472→0.8931? Because multi-view fusion at boundaries also judges some "in-truncation-band" voxels occupied (multi-view accumulation thickens edges), causing predicted mask slightly larger than true mask; numerator (intersection) rises while denominator (predicted mask) rises more, precision slightly falls. This is a structural tradeoff between coverage and precision, not a bug.

Appendix E 3DGS-lite Fitting Process and Alpha Compositing

`render_splat`'s front-to-back alpha compositing: for pixel p , sort gaussians by depth, accumulate transmittance $T = \prod_i (1 - \alpha_i)$, where $\alpha_i = o_i \exp(-\frac{1}{2} d_i^2)$, d_i the pixel-to-gaussian-projected-center Mahalanobis distance (axis aligned). Output color $C = \sum_i w_i c_i$, depth $Z = \sum_i w_i z_i$, $w_i = T \alpha_i$. Mask takes $1 - T > 0.30$. Cold-started from TSDF occupied voxels, gaussian centers initially near object surfaces, fitting only adjusting color and scale; within Adam 40 steps total loss 1.7217→0.5694, depth L1 0.5065→0.1328—showing cold-start positions already reasonable, fitting mainly converging depth alignment.

Coverage 1.0 but IoU 0.7285 means: alpha compositing produces nonzero contribution for almost all predicted pixels (mask filled), but axis-aligned gaussians under oblique views cannot precisely fit the sphere's ellipsoid silhouette, predicted vs true mask intersection-over-union stops at 0.73. Depth MAE 0.2525 higher than TSDF's 0.0924, because splat uses small gaussians to softly merge depth, while TSDF directly gives hard occupied surfaces. This is the prototype-level "rendering flexibility vs geometric accuracy" tradeoff.

Appendix F Sim-to-Real Mismatch Construction Details

`sim2real_gap`'s mismatch explicitly constructed: planner-seen obstacle positions overlaid $\mathcal{N}(0, 0.12^2)$ offset, radius shrunk to truth's $0.9\times$; while the "real" world obstacle positions overlaid another $\mathcal{N}(0, 0.10^2)$ offset, radius overlaid $\mathcal{N}(0, 0.04^2)$. This means the planner has systematic bias in both obstacle position and size. Naive planning directly generates trajectories with this biased estimate; robust planning further inflates estimated radius by `margin=0.22`, equivalent to reserving an "uncertainty buffer band" in planning geometry. Result: naive planning half the worlds collide into real obstacles (success 0.5, mean collisions 0.5), while the buffer wraps real obstacles inside the inflated estimate spheres, planned trajectories detour around inflated spheres, hence around real obstacles (success 1.0, collisions 0.0).

This result's boundaries must be clear: first, margin 0.22 a specific value under single main obstacle, fixed noise level, not universally optimal; second, too large a margin makes passable corridors disappear, planning fails; third, real-world mismatch is not only position/radius gaussian noise but also shape, friction, dynamics differences, the latter not fully coverable by geometric inflation. Hence this paper states H3 as "under constructed mismatch, positive margin improves success and eliminates

collisions," not "margin is Sim-to-Real's universal solution." ϵ critical curve (success vs ϵ from 0 first rises then falls inflection) is the next core measurement.

Appendix G Dialogue with External World Model Claims

External reports say novel-view prediction and next-token prediction same level, say 3DGS renders fast but struggles with dynamics, say Real-to-Sim can compress real environment reconstruction cost. This paper's CPU prototype provides three "explicit geometric level" corroborations: first, even without neural networks, only multi-view TSDF fusion, novel-view coverage rises 0.75→0.996—showing "view geometric context" indeed novel-view prediction's hard constraint, not purely a learning phenomenon; second, 3DGS-lite static rendering usable but silhouette accuracy limited, confirming "splat renders fast but expressiveness limited"; third, Real-to-Sim reconstruction radius vs truth difference about 0.013, end-to-end control 100% success, showing on synthetic data the "observe→reconstruct→plan" link works. But these all `cpu-proto` corroborations, not reproductions or endorsements of external systems.

Appendix H Measurement Methodology and Metric Definitions

This paper holds metrics frozen before data collection and mechanically computed by code. Four reconstruction metrics defined, all from `depth_metrics`: silhouette IoU predicted vs true mask intersection-over-union; coverage intersection pixels divided by true mask pixels (measuring "how much of the true surface predicted"); precision intersection pixels divided by predicted mask pixels (measuring "how much predicted is correct"); depth MAE only computed on overlapping pixels (avoiding no-overlap pixels biasing error). This distinction crucial: single view precision high (0.947) but coverage low (0.746), reporting only precision would mislead that single view good; precisely the coverage metric exposes the "missed surface" problem.

Sim-to-Real's two metrics likewise mechanical: success judged by the real proxy world's `env.success` (reach target and zero collisions), mean collisions the eight worlds' collision count average. Note naive planning's "mean collisions 0.50" not "each world collides 0.5 times," but eight worlds' collision count average—under contact terminal, one collision terminates, hence that value actually reflects "what fraction of worlds had a collision." This caliber consistent with success 0.50: half worlds collided, half not.

Real-to-Sim's "radius reconstruction accuracy" needs special note. Report's `planning_obstacle` is `[[0.902,-0.506,0.0],0.703]`, i.e. reconstructed obstacle center (0.902,-0.506), z=0 plane section radius 0.703; `true_plane_obstacles=2` means truth has two plane-intersecting obstacles. The one-pager and document's " $r \approx 0.70$ " is a qualitative statement about this one main obstacle radius 0.703, **not Pearson correlation coefficient**. With only two obstacle clusters, only one becoming the planning main obstacle, sample size insufficient to support any correlation significance conclusion. This paper in Table 3 and Section 6.3 states this caliber, not writing it as a "correlation $r \approx 0.70$ " statistical finding.

Appendix I Main Loop Pseudocode (corresponding to source)

Novel-view prediction main link corresponds to `spatial_novelview_demo.py` and `fusion.py`:

```

Input: scene, training view set (single uses 1, multi uses 8), held-out views 6
1. for cam in training views:
2.     d, mask = render_view(cam, scene)          # analytic intersection true depth
3.     ctx.integrate(d, mask, cam)                # accumulate TSDF
4. for cam in held-out views:
5.     pred_d, pred_mask = novel_view_depth(cam, ctx) # step along ray hit occupied
6.     gt_d, gt_mask = render_view(cam, scene)
7.     metrics[cam] = depth_metrics(pred_d, pred_mask, gt_d, gt_mask)
8. aggregate six held-out views' IoU/coverage/precision/depth_mae

```

3DGS-lite and transfer main link corresponds to `fit_splat` and `sim2real_gap`:

```

1. ctx = fuse_views(training noisy views)
2. splat = init_splat_from_context(ctx, scene, n_gaussians=150)
3. for it in range(40):                                # Adam
4.     loss = splat_loss(splat, training views, scene)
5.     loss += 1e-3*splat.opacity.abs().mean()
6.     loss.backward(); opt.step(); clamp(colors/opacity/scales)
7. held-out evaluation: splat_novelview_report(splat, 4 new views, scene)
8. Real2Sim: extract_obstacle_spheres(ctx) -> gate task -> hybrid control
9. Sim2Real: construct biased estimate vs real perturbation worlds
10. naive: biased estimate direct plan execute -> success/collisions
11. robust: estimate radius inflation margin=0.22 -> success/collisions

```

Appendix J Extended Analysis Directions

First, view count scan. Currently only 1 vs 8. Expected on view count {1,2,4,8,16} coverage monotonically rises and saturates at some point (after about 8 views marginal gain tends zero), while precision may first rise then fall. This scan is H1's key upgrade from "two-point comparison" to "scale curve," not run [RESULT NEEDED: view count saturation curve].

Second, view layout ablation. Ring surround vs front-facing vs random views, coverage differences on occluded objects should be large. This decomposes "multi-view" from a black box into the continuous variable "baseline distance between views."

Third, noise level scan. As `depth_sigma` and `dropout` rise, TSDF weighted fusion automatically lowers low-confidence observations' influence (weighted average), should measure reconstruction fidelity robustness curves to noise, comparing with 3DGS-lite differentiable fit robustness.

Fourth, ϵ critical curve. Scanning margin 0 to 0.5, expected success first rises then falls (too large margin no feasible path), inflection the "minimum sufficient margin." This is the core measurement upgrading H3 to a predictable design rule.

Fifth, standard rendering metrics. Implementing PSNR/SSIM/LPIPS to be comparable with 3DGS literature; currently only silhouette IoU and depth MAE, not compared with paper-level metrics.

All above to-be-executed work, this paper not pre-filling results.

Appendix K Reproducibility and Author Statements

This link's random sources all driven by fixed-seed `torch.Generator`: noisy observation, camera jitter, 3DGS initialization sampling all reproducible on the same tag. Rendering analytic ray intersection, no external dependencies. All numbers rerunnable by Appendix A commands on CPU to byte-identical JSON. External Atlas claims isolated in the `external_reference` field, entering no aggregate computation. This paper generates no fabricated literature, no invented p-values or CIs; Real2Sim's small sample and Sim2Real's synthetic world boundaries explicitly stated in Sections 7,8. AI-assisted drafting note at end.

Appendix L Summary

This paper on the CPU geometric prototype proves three things: multi-view spatial context raises novel-view reconstruction coverage about 0.75→0.996 and eliminates direction dependence; minimal 3DGS-lite reaches usable silhouette IoU (0.7285) on held-out views; while under Sim-to-Real mismatch, naive planning half fails, after 0.22 explicit safety margin success and collisions become 1.0 and 0.0. This "negative result retained + positive margin effective" contrast has more methodological value than simply reporting "our method is good." Next after GPU and real sensors unlock, using the same metric framework to test whether these explicit geometric conclusions hold on learned representations and the real world.

Appendix M Further Expansion of Related Work

Novel-view synthesis roughly went through three stages. Early using explicit geometry (stereo, visual hull, voxel reconstruction) to recover 3D surfaces from multiple views; neural radiance fields representing scenes as continuous density and color fields, optimized by differentiable rendering, greatly improving occlusion handling and novel-view fidelity, but training slow, rendering needing per-point sampling; 3D Gaussian Splatting combining explicit ellipsoid primitives with differentiable EWA rasterization, achieving real-time high-quality rendering. This paper's TSDF belongs to the first-stage explicit geometry, 3DGS-lite borrowing the third-stage primitive idea but maximally simplified. Why placing explicit TSDF and simplified splat in one paper is because they answer different questions: TSDF isolates "multi-view geometric fusion's own gain," splat-lite answers "how much to cold-start an optimizable representation from fused occupied points."

Sim-to-Real's core difficulty is the "reality gap." Mitigations include domain randomization (applying wide-range random perturbations in simulation to make policies robust to real perturbations), system identification (fitting simulation parameters to real data), domain adaptation (aligning simulation and real feature distributions). This paper's "radius inflation margin" is domain randomization's simplest form at geometric planning: not changing policy learning, only reserving uncertainty bands in planning geometry. Its advantage interpretable, measurable, disadvantage only covering geometric position/size uncertainty, not dynamics and contact uncertainty. Combining it with policy-layer domain randomization is a future direction.

Real-to-Sim reverses: reconstructing a simulatable environment from real sensors. This paper uses noisy depth + pose jitter to proxy real capture, TSDF reconstruction then extracting obstacle spheres, then generating navigation tasks. This link's value is making "how sensor noise propagates to planning geometry" measurable: how depth noise and dropped points affect occupied voxel weights, how pose jitter makes reconstructed spheres deviate from truth, finally how it appears in Sim-to-Real mismatch.

This paper reports end-to-end success, but single obstacle, single scene, not yet a system-level Real-to-Sim conclusion.

Appendix N Position on the "AI-Complete Foundational Task" Claim

External reports place novel-view prediction alongside next-token prediction as an AI-complete foundational task. This paper takes a restrained attitude to this grand claim. Our evidence only supports a weaker proposition: **on synthetic sphere scenes, multi-view geometric context is novel-view prediction's hard constraint, and its gain explicitly measurable.** As for "does novel-view prediction like next-token prediction generalize all intelligence," neither answerable by this paper's data nor intended claim. Drawing a clear line between explicit geometric prototype findings and this philosophical claim is part of this paper's evidence discipline. Readers should not infer any conclusion about general world model ability from this paper's 0.8897 IoU.

Appendix O Metric Cross-Verification and Recomputation Record

To ensure numbers traceable, this paper recomputes key differences. Multi vs single coverage difference $0.9957 - 0.7456 = 0.2501$, IoU difference $0.8897 - 0.7147 = 0.1750$, depth MAE reduction ratio $(0.3186 - 0.0924) / 0.3186 \approx 0.71$, i.e. multi-view reduces depth error about 71% (to about $0.29\times$, i.e. a bit over one third). Loss history recomputation: iter0 to iter30 total loss decline $1.72170 \rightarrow 0.56940$, about 67%; depth L1 decline $0.50653 \rightarrow 0.13277$, about 74%, the main total loss decline source. Sim-to-Real naive planning mean collisions 0.50 and success 0.50 under contact terminal mutually confirm (eight worlds four collided, four not). Real-to-Sim main obstacle reconstructed radius 0.703 vs true plane section 0.69 difference about 0.013, relative error about 1.9%, but sample single main obstacle, no statistical inference. The above recomputations all directly from JSON fields, introducing no external assumptions.

Appendix P Limitations and Follow-up Experiment Checklist Summary

This paper's limitations reducible to five: fixed synthetic sphere scene, no texture no real camera, 3DGS-lite no rotated covariance and EWA, Sim-to-Real synthetic perturbation proxy not real hardware, Real-to-Sim only single main obstacle gate. Follow-up experiments by priority: first, expand to no fewer than thirty random scenes and run view count, layout, noise level grid scans, reporting paired IoU tests and CIs; second, scan robustness margin ϵ critical curves, giving "minimum sufficient margin" design rules; third, implement PSNR/SSIM to connect with 3DGS literature; fourth, after GPU unlock run full 3DGS and MuJoCo-MJX contact transfer; fifth, after real depth camera calibration retest Real-to-Sim. Before these experiments complete, all this paper's conclusions `cpu-proto` point estimates, no SOTA claim, no real-hardware generalization claim. This restraint both scientific honesty and the premise making follow-up work falsifiable.

Must stress, this paper deliberately reports negative results (naive Sim-to-Real only half success, mean collisions 0.50) alongside positive results (after margin injection all success, zero collisions), not only showing success gains. This contrast itself a methodological position: only simultaneously seeing "what happens without safety margin" can one truly understand the margin's value, and avoid packaging the method as unconditionally effective. So on synthetic worlds, even more so on the real world.

Author Intended Statements

- **Target journal/conference:** current evidence strength (CPU geometric prototype, fixed scene) suits 3DV/BMVC/WACV/ICRA or CVPR workshop short paper; after evidence upgrade (GPU, multi-scene, standard metrics) able to submit ICRA/IROS, ISPRS Journal, IEEE RA-L, CVIU. Partitions/impact factors all [to verify], verified online item by item before submission with verification dates.
- **Pre-registration plan:** Section 5's "view count $\times$ layout $\times$ noise" grid and ϵ critical curve pre-registration draft; registering before sample expansion, fixed seed table and analysis code released with tag; main metrics (held-out IoU/coverage), guards (depth MAE, collision rate) and paired tests frozen before data collection.
- **Data and code availability:** engine open-sourced under Apache-2.0, code tag v7.5.0 (commit 1a71270); report sha256 in Appendix A/B, tests7/test_v738/v739 provided with repo.
- **AI use statement:** this paper's draft assisted by an AI assistant based on the repo's real source and reports7 measured data, all numbers back-sourced to spatial_novelview_demo.json and splat_transfer_demo.json; external Atlas claims marked unverified, not cross-proven with this repo; unmeasured ϵ critical curves, PSNR/SSIM etc. marked [RESULT NEEDED]; references filled after academic search verification.